

DETAILED DESCRIPTION OF THE INVENTION

Future Expansion and Adaptive Modeling Architectures

In various embodiments, the penile anatomical geometry profiling system may incorporate adaptive modeling frameworks configured to refine reconstruction accuracy and enhance predictive analytics over time. The geometry reconstruction engine may utilize machine learning algorithms trained on aggregated geometry datasets to improve interpolation accuracy, compensate for sparse sampling regions, or predict surface contours between measured intervals.

Adaptive modeling systems may analyze historical user measurements to detect consistent structural baselines and identify statistically significant deviations across sessions. Predictive modeling may evaluate trends in longitudinal length stability, radial distribution changes, curvature variation, or volumetric fluctuation across time-series datasets. Such predictive outputs may be expressed as confidence intervals, stability indices, or deviation thresholds.

In certain embodiments, machine learning models may be trained to classify geometry profiles into multidimensional structural categories using supervised or unsupervised learning techniques. Feature extraction routines may identify latent structural relationships among parameters including taper gradients, curvature vectors, symmetry scores, and proportionality ratios. Adaptive weighting functions may adjust parameter influence based on population-level statistical modeling.

The system may further implement real-time adaptive sampling strategies in which sensor acquisition density is dynamically modified based on preliminary reconstruction outputs. Regions exhibiting higher curvature or irregularity may trigger increased sampling frequency, while uniform regions may utilize reduced sampling density to optimize efficiency.

In some implementations, federated learning architectures may be employed such that device-level models are trained locally using anonymized structural parameters and aggregated model updates are transmitted without exposing raw geometry data. This approach may enable global model refinement while preserving user privacy.

The adaptive and predictive modeling architectures described herein are not limited to any specific machine learning technique, training protocol, data aggregation method, or computational framework, provided that the system maintains the capability of generating, refining, or contextualizing a user-specific penile anatomical geometry profile derived from synchronized time-of-flight and motion-based measurements.